

Hong Chul Nam

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EDUCATION

New York University

Doctor of Philosophy in Data Science (CDS/Courant Institute)

New York City, USA

Aug 2026 – Present

ETH Zurich

Master of Science in Electrical Engineering and Information Science

Zurich, Switzerland

Jan 2022 – Mar 2026

California Institute of Technology

Visiting Researcher

Pasadena, USA

Oct 2023 – Sept 2024

ETH Zurich

Bachelor of Science in Electrical Engineering and Information Science

Zurich, Switzerland

Sep 2018 – Sep 2023

St. Anselm's Abbey School

High School

Washington DC, USA

Sep 2014 – Sep 2016

EXPERIENCE

Visiting Student

Harvard, Multiply Labs

Jan 2026 – Present

Remote

- Supervisor: Dr. Haonan Chen, Dr. Jiuguang Wang, Prof. Yilun Du
- Working on extending energy-based models into Lagrangian formulation for developing energy-based world models for video generations and policy learning

AI Researcher

Zenithon AI

Dec 2025 – Present

Remote

- One of AI researchers working on building a billion-parameter foundation model for plasma physics using transformer-based neural operators to accelerate physical simulation of nuclear fusion

AI Researcher

Korea Military Academy

Oct 2024 – Present

Seoul, Republic of Korea

- Supervisor: Sunil Hwang, Prof. Hyun Kwon
- Mandatory military service (1.5 years)
- Participated (top 2 out of 270 teams) in the public competition for AI-based automatic document generation using VLM for in-home elder care work
- Built an autonomous surveillance system using real-time object detection with VLM for the Korean army
- Achievements: publication in CoRL Workshop 2025, second-place in competition for document automation for in-home elder care works

Student Researcher

Anima's Lab, California Institute of Technology

Oct 2023 – Present

Pasadena, USA

- Funding: Caltech Research Funding
- Supervisor: Dr. Julius Berner, Dr. Xi Deng, Prof. Anima Anandkumar
- Solved high-dimensional Poisson equations using stochastic reformulation via Walk-on-Spheres (WoS) and neural network in a general framework called Neural Walk-on-Spheres (NWoS)
- Extending self-supervised learning methods (VICReg, MAE, JEPa) to functions with neural operators
- Collaborating with researchers from NVIDIA, University of Chicago, Argonne National Laboratory and Caltech to improve weather modeling with self-supervised learning
- Collaborating with a researcher from Purdue University to extend Neural Walk-on-Spheres (NWoS) to neural operators (WoS-NO)
- Co-supervising a student on neural operator learning for path planning via screened Poisson equations and WoS.
- Generalizing random walk schemes on training data generated from expensive mesh-based solvers for learning transition kernel (i.e. Green's function) to enable generalization of stochastic solvers into nonlinear PDEs
- Achievements: publication in ICML 2024, submission to ICLR 2026

AI Researcher

Alsemy Inc.

Jan 2022 – Dec 2024

Seoul, Republic of Korea

- Supervisor: Chanwoo Park, Dr. Ye Sle Cha, Dr. Jongwook Jeon, Hyunbo Cho
- Accelerated modeling of variability in transistor devices under process variations using generative models and importance sampling
- Implemented neural operator-based foundation models and reinforcement learning methods for transistor device modeling and circuit optimization

- Created the first function-level anomaly detection neural operator for detecting anomaly device simulations
- Created the device simulation augmentation method by training a function-level autoencoder and latent diffusion model to improve the diversity of large-scale training data
- Proposed a neural ODE-based compact model for transient self-heating simulation of transistor devices
- Achievements: publications in SISPAD 2023/2024, NeurIPS 2023 Workshop, ICMC 2025, IEEE Electron Device Letters Submission; full simulation pipeline for variability analysis, DTCO, anomaly detection and self-heating

Research Assistant

Sep 2021 – Jul 2023

SAFARI Lab, ETH Zurich

Zurich, Switzerland

- Supervisor: Konstantinos Kanellopoulos, Prof. Onur Mutlu
- Created a neural network-based page table walk cost estimator to selectively place evicted L2 TLB page tables into L2 data cache
- Introduced Merkle trees in page tables to enforce efficient security checks during each intermediate walk of page table walks to verify the integrity of page tables under security attacks
- Contributed to CUDA programming introductory codes for a heterogeneous computing course
- Achievements: publication in MICRO 2023; extension of SniperSim to support neural network; extension of Gem5 and SniperSim to support Merkle trees in page tables

Research Intern

Sep 2021 – Apr 2022

Swiss Center for Electronics and Microtechnology (CSEM)

Zurich, Switzerland

- Supervisor: Dr. Andrea Bonetti
- Developed a low-power MFCC feature extractor using SystemVerilog with UVM verification for a speech signal AI accelerator
- Explored structural changes in hardware as well as neural network architecture for achieving ultra-low power consumption
- Achievements: implementation of MFCC module in SystemVerilog code with verification; optimization of hardware/software implementation of the block

Research Engineer

Sep 2020 – Sep 2021

ASL Lab, ETH Zurich

Zurich, Switzerland

- Funding: Swiss military, ETH Zurich
- Supervisor: David Rohr, Dr. Nicholas RJ Lawrance, Prof. Roland Siegwart
- Developed an autonomous firefighting VTOL drone for wildfire surveillance from scratch
- Achievements: development of VTOL drone with surveillance system

Research Assistant

Jun 2015 – Jul 2015

Vanderver Laboratory, Children's National Hospital

Washington DC, USA

- Funding: Summer internship stipend
- Supervisor: Dr. Asako Takanohashi, Prof. Adeline Vanderver
- Worked on Aicardi Goutieres Syndrome using PCR, cell culture, western blot, and immunohistochemistry approaches
- Achievements: learning cellular biology and all techniques for growing cells and treating drugs

PROJECTS

Self-supervised Learning for Few-Step Models

Jun 2025 – Present

- Collaborator: Sunil Hwang, Junseob Kim, Jaehyeong Jo
- Quantifying the loss of sample diversity in few-step models such as the consistency model and meanflow
- Introducing semantic information with alignment to improve the diversity

Debiasing the Classifier-Free Guidance due to Nulltoken

Dec 2024 – Present

- Collaborator: Joochul Lee
- Proved quantitatively the bias in probability distributions due to null token for approximating the unconditional distribution in conditional flow matching models and working on removing the bias

Q-Guided Flow Q-Learning

Dec 2024 – Sep 2025

- Supervisor: Prof. Hyun Kwon
- Developed a decoupled actor-critic framework combining flow-based policy learning and value-guided correction for stable and expressive offline reinforcement learning

Continuous Control of Robot Arm

Jan 2023 – Jun 2023

- Supervisor: Bhavya Sukhija, Lenart Treven, Prof. Andreas Krause
- Implemented robot API for RL in the real world for the UR5 robot

- Implemented and tested offline model based RL methods on the robot
- Achievements: implemented software interface to control UR5 robot arm; verified RL methods in the robot

Autonomous Driving Mini-car

Sep 2020 – Jan 2021

- Supervisor: Dr. Michele Magno, Prof. Luca Benini
- Developed a real-time autonomous driving model using a Sony microcontroller and an RC car with external camera modules for CNN-based lane detection
- Achievements: demonstrated the highly accurate and functional autonomous driving system with Sony microcontrollers using CNN-base architectures

PUBLICATIONS

- **Hong Chul Nam***, Jeonghwan Cheon*, Jaemo Shin, Hyun Kwon. "Energy-Based Operator Learning in Function Space". Submission: Conference on Neural Information Processing Systems (NeurIPS 2026)
- Hrishikesh Viswanath*, **Hong Chul Nam***, Julius Berner, Anima Anandkumar, Aniket Bera. "Operator Learning Using Weak Supervision from Walk-on-Spheres". In: arXiv preprint arXiv:2603.01193 (2026).
- Eunki Joung, **Hong Chul Nam**, Geonhwi Hwang, Moohyun Lee, Hyun Kwon. "Making Visible, Making Invisible: How an AI Scribe Reshapes Documentation Authority in Social Work". In: Trustworthy AI for Good (AI4GOOD) Workshop@ ICML 2026.
- Yejun Jang*, **Hong Chul Nam***, Jeong Min Park*, Gimin Bae, Hyun Kwon. "Q-Guided Flow Q-Learning". In: CoRL 2025 Workshop RememberRL
- Premkumar Vincent*, Yeongwoo Nam*, Kyungmin Kim*, **Hong Chul Nam**, Hyunseok Whang, Donghyun Jin, Jongwook Jeon, Chang-Sub Lee, Jihun Park, Ye Sle Cha, Hyunbo Cho. "Large Pre-trained Model Approach for Efficient Design Technology Co-Optimization". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).
- **Hong Chul Nam***, Tae Il Oh*, Ye Sle Cha, Jongwook Jeon, Hyunbo Cho. "FuncFlow: A Generative Neural Operator Using Diffusion Model for Simulation Augmentation". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).
- **Hong Chul Nam***, Julius Berner*, Anima Anandkumar. "Solving Poisson Equations using Neural Walk-on-Spheres". In: Forty-first International Conference on Machine Learning (ICML 2024).
- Tae Il Oh*, **Hong Chul Nam***, Chanwoo Park, Hyunbo Cho. "FuncAnoDe: A Function Level Anomaly Detection in Device Simulation". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2024).
- Konstantinos Kanellopoulos, **Hong Chul Nam**, Nisa Bostanci, Rahul Bera, Mohammad Sadrosadati, Rakesh Kumar, Davide Basilio Bartolini, Onur Mutlu. "Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources". In: Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO 2023).
- Chanwoo Park*, **Hong Chul Nam***, Jihun Park, Jongwook Jeon. "FlowSim: An Invertible Generative Network for Efficient Statistical Analysis under Process Variations". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2023).
- **Hong Chul Nam**, Chanwoo Park. "NPC-NIS: Navigating Semiconductor Process Corners with Neural Importance Sampling". In: NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.

AWARDS

Second place (top 2 / 270 teams): Competition for in-home elder social care (Aug 2025)

Third place: Defense Technology Startup Competition, Korean Army (May 2025)

Fourth place: Army Startup Competition (Mar 2025)

TECHNICAL/LINGUISTIC SKILLS

Linguistics: English, Mandarin, Korean, German

Languages: Java, Python, C/C++, Verilog/SystemVerilog

ML Libraries: PyTorch, TensorFlow, JAX

Computer Simulators: Gem5, SniperSim